

SDR Monitoring Spectrum System

Predesign Topics Review

DesignSensorSpectrum

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System Goal

Build a monitoring node around an embedded compute unit, an SDR front-end, controlled RF switching, and optional remote connectivity plus geolocalization.

- **SDR front-end:** Ettus USRP B200mini.
- **Compute unit options:** Jetson Nano or Raspberry Pi 5.
- **RF switching:** HMC849 SPDT for branch selection.
- **Remote node functions:** LTE backhaul and GNSS through SIM7600G-H.
- **Critical engineering axes:** power integrity, impedance control, thermals, and RF input protection.

Power Tree: How to Do It

- ❶ **List all loads:** compute board, SDR, RF switch, modem, fan, storage, and optional active antennas.
- ❷ **Separate power domains:** noisy digital loads, RF/control loads, and high-peak loads such as LTE.
- ❸ **Size worst-case current:** streaming SDR + modem TX + fan + storage is the design point, not idle mode.
- ❹ **Choose regulation:** use the official SBC input rail as the root, then derive clean 5 V / 3.3 V branches as needed.
- ❺ **Add protection:** fuse or current limit, reverse-polarity protection, filtering, TVS where appropriate, and ground strategy.
- ❻ **Validate margins:** check cable drop, peak current, boot current, and thermal rise under simultaneous load.

Applied to This Design

Root input → SBC rail → USB 3.0 SDR branch + RF switch control rail + LTE/GNSS branch + cooling branch.

SDR Internal Specs: Hardware and Software

Hardware View

- AD9364-based RF transceiver.
- **70 MHz to 6 GHz** operating range.
- **12-bit** ADC/DAC.
- Up to **61.44 MS/s**.
- **J1 TRX** shared TX/RX and **J2 RX2** dedicated RX.
- External **10 MHz / PPS** support on J3.
- Maximum safe RF input: **-15 dBm**.

Software View

- Managed through **UHD**.
- Key parameters: `sample_rate`, `bandwidth`, `RX/TX gain`, `AGC`, `master_clock_rate`.
- Analog LPF and digital filters are configured through UHD.
- IQ imbalance and DC offset correction are already built into the transceiver/UHD stack.
- System bottlenecks are usually host CPU, USB throughput, and buffer overrun/underrun handling.

Impedance Matching: 50 Ω RF vs 75 Ω TDT

50 Ω Domain

- SDR ports, SMA cables, attenuators, switch modules, filters, and lab accessories.
- This should be the controlled impedance of the internal RF chain.
- Keep unused TX or test ports terminated with **50 Ω** .

75 Ω TDT Domain

- Typical for terrestrial TV antennas, splitters, and domestic coax runs.
- If the system connects to TDT infrastructure, adapt the transition into the SDR chain.
- For calibrated work, use a 75 $\Omega \leftrightarrow$ 50 Ω pad, transformer, or matching section.

Practical Rule

A direct 75 Ω to 50 Ω connection often still works, but it is a mismatch. For repeatable measurements, define one controlled adaptation point and keep the rest of the chain at 50 Ω .

Compute Unit: Expected Resources

Platform	Official Core Specs	Predesign Reading
Jetson Nano	4x Cortex-A57 @ 1.43 GHz, 4 GB LPDDR4, 128-core Maxwell GPU, 472 GFLOPS, 5–10 W	Better when the node also needs CUDA or AI inference near the SDR pipeline. Limited CPU compared with newer SBCs.
Raspberry Pi 5	4x Cortex-A76 @ 2.4 GHz, LPDDR4X options from 1 GB to 16 GB, VideoCore VII, 2x USB 3.0, GbE, 5 V / 5 A recommended	Stronger general CPU platform for host-side DSP, logging, and orchestration if GPU acceleration is not the main requirement.

Sizing Guideline

For plain spectrum monitoring and control, Raspberry Pi 5 is the stronger CPU-first option. For AI-assisted classification, camera fusion, or CUDA workflows, Jetson Nano keeps an advantage at the platform level.

Thermal Throttling

- Thermal throttling is not only a CPU problem; it impacts **sample stability**, USB reliability, modem uptime, and enclosure behavior.
- Main heat sources in this design are:
 - compute SoC,
 - LTE modem during transmission,
 - storage and power stages,
 - confined airflow around the SDR host.
- Raspberry Pi 5 official operating temperature is **0°C to 70°C**; the enclosure should be designed to stay below thermal reduction zones.
- Jetson Nano power modes are low, but sustained AI or video tasks still need heatsinking and airflow.
- Predesign mitigation: fan header budget, vent path, modem spacing, thermal pads/heatsinks, and logging of temperature under real SDR streaming load.

Undervoltage

- Undervoltage causes unstable USB peripherals, SDR disconnects, sample drops, storage errors, and LTE modem resets.
- Worst cases appear during **boot**, **USB 3.0 traffic**, and **SIM7600 TX current peaks**.
- Raspberry Pi 5 official guidance:
 - it can boot from **5 V / 3 A**,
 - but high-power peripheral current is then restricted,
 - **5 V / 5 A** gives more USB current budget and more system margin.
- Predesign rules:
 - keep input cables short and low resistance,
 - avoid powering modem peaks from a weak shared branch,
 - leave current margin for fan and storage,
 - validate supply sag with all subsystems active at once.

Internet or Node Connection

Lab / Fixed Node

- Gigabit Ethernet for stable remote access.
- SSH, rsync, VPN, or web dashboards.
- Easier for continuous logging and software updates.

Field / Mobile Node

- SIM7600G-H provides LTE backhaul.
- USB is the recommended integration path with Jetson-class hosts.
- Good for remote campaigns, telemetry, and alerting.

Node Architecture Guideline

Do local capture and local reduction first. Send metadata, alarms, and summarized spectra over the network; send raw wideband IQ only when the link budget allows it.

- The **HMC849 SPDT** is the predesign choice for switching between two RF branches.
- Current architecture idea:
 - USRP **RX2** connected to the common switch port,
 - output A and output B connected to two antennas or two filter paths,
 - GPIO from the compute unit selects the active branch.
- Typical uses:
 - antenna diversity,
 - band-select filtering,
 - protected bypass topologies.
- The switch is **not** an attenuator, limiter, or surge protector.
- Predesign checks: logic threshold compatibility, insertion loss, isolation, RF power limit, and board-level grounding.

- The **SIM7600G-H** adds **GPS, BeiDou, GLONASS, Galileo, and QZSS** support.
- Geolocation is useful for:
 - tagging spectrum captures by position,
 - mobile drive-test style campaigns,
 - asset tracking of deployed nodes,
 - correlating events with location and route.
- Integration options:
 - GNSS queried over AT commands,
 - stored in local logs with timestamp and node ID,
 - forwarded to a server together with alarms or summary measurements.
- Important distinction: GNSS gives position; precise SDR coherence still benefits from a dedicated **10 MHz / PPS** timing strategy if future synchronization is required.

- Keep the internal RF chain at **50 Ω** and define explicit adaptation for TDT-style 75 Ω inputs.
- Size power around **peak simultaneous load**, not around nominal board values.
- Pick **Raspberry Pi 5** for CPU-centric SDR nodes, or **Jetson Nano** when GPU/AI integration is part of the roadmap.
- Treat thermal throttling and undervoltage as first-order system risks.
- Use LTE and GNSS as node services, not as substitutes for RF timing or front-end protection.